Chapter 31 Review

Campbell Biology, AP Edition, 12e

Chapter 31 Review

Summary of Key Concepts

Many lichens are very sensitive to changes in atmospheric chemistry, and lichen die-offs can serve as environmental indicators of air pollution. Which of the mutualistic organisms that make up lichens (fungi or algae/cyanobacteria) is most likely to be affected by changes in the air, causing the entire lichen to die? **(BIG IDEAS 2 & 4)**

Concept 31.1: Fungi are heterotrophs that feed by absorption

- All **fungi** (including decomposers and symbionts) are heterotrophs that acquire nutrients by absorption. Many fungi secrete enzymes that break down complex molecules.
- Most fungi grow as thin, multicellular filaments called **hyphae**; relatively few species grow only as single-celled **yeasts**. In their multicellular form, fungi consist of **mycelia**, networks of branched hyphae adapted for absorption. Mycorrhizal fungi have specialized hyphae that enable them to form a mutually beneficial relationship with plants.

How does the morphology of multicellular fungi affect the efficiency of nutrient absorption?

Concept 31.2: Fungi produce spores through sexual or asexual life cycles

- In fungi, the sexual life cycle involves cytoplasmic fusion (**plasmogamy**) and nuclear fusion (**karyogamy**), with an intervening **heterokaryotic** stage in which cells have haploid nuclei from two parents. The diploid cells resulting from karyogamy are short-lived and undergo meiosis, producing genetically diverse haploid **spores******.
- Many fungi can reproduce asexually as filamentous fungi or yeasts.

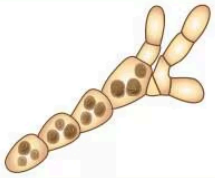
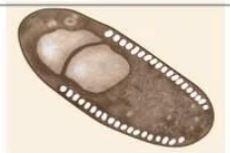
DRAW IT *Draw a generalized fungal life cycle, labeling asexual and sexual reproduction, meiosis, plasmogamy, karyogamy, and the points in the cycle when spores and the zygote are produced.*

Concept 31.3: The ancestor of fungi was an aquatic, single-celled, flagellated protist

- Molecular evidence indicates that fungi and animals diverged over a billion years ago from a common unicellular ancestor that had a flagellum. However, the oldest fossils that are widely accepted as fungi are 440 millionyears old.
- Chytrids and other basal fungal lineages have flagellated spores.
- Fungi were among the earliest colonizers of land; fossil evidence indicates that these colonizers included species that were symbionts with early plants.

Did multicellularity originate independently in fungi and animals? Explain.

Concept 31.4: Fungi have radiated into a diverse set of lineages

Fungal Phylum	Distinguishing Features
Cryptomycota (**cryptomycetes**)	Parasites with flagellated spores 
Microsporidia (**microsporidians**)	Parasitic cells that form resistant spores 
Chytridiomycota (**chytrids**)	Flagellated spores
Zoopagomycota (**zoopagomycetes**)	Resistant zygosporangium as sexual stage
Mucuromycota (**mucuromycetes**)	Include fungi that form arbuscular mycorrhizae with plants
Ascomycota (**ascomycetes**)	Sexual spores (ascospores) borne internally in sacs called asci; vast numbers of asexual spores (conidia) produced
Basidiomycota (**basidiomycetes**)	Elaborate fruiting body (basidiocarp) containing many basidia that produce sexual spores (basidiospores)

The table is as follows:

Fungal phylum	Distinguishing features
Cryptomycota (cryptomycetes)	Parasites with flagellated spores. An image of a fungus invaded by parasitic spores of Cryptomycetes.
Microsporidia (microsporidians)	Parasitic cells that form resistant spores. A parasitic cell of Microsporidia, which forms resistant, elliptical spores.
Chytridiomycota (chytrids)	Flagellated spores. An image of a flagellated chytrid.
Zoopagomycota (zoopagomycetes)	Resistant zygosporangium as sexual stage. An image illustrating the zoosporangium of Zoopagomycetes, in a sexual stage.
Mucuromycota (mucuromycetes)	Include fungi that form arbuscular mycorrhizae with plants. An image illustrating a fungus forming arbuscular mycorrhizae with plants, belonging to Mucoromycetes.
Ascomycota (ascomycetes)	Sexual spores (ascospores) borne internally in sacs called asci; vast numbers of asexual spores (conidia) produced). An image illustrating the release of sexual spores (ascospores) from the asci through the ascocarp of an ascomycete, bread mold (Neurospora).
Basidiomycota (basidiomycetes)	Elaborate fruiting body (basidiocarp) containing many basidia that produce sexual spores (basidiospores). A part of the life cycle of a basidiomycete showing the fruiting body basidiocarp releasing basidiospores from a basidium.

DRAW IT *Draw a phylogenetic tree of the major groups of fungi.*

Concept 31.5: Fungi play key roles in nutrient cycling, ecological interactions, and human welfare

- Fungi perform essential recycling of chemical elements between the living and nonliving world.
- ****Lichens** are highly integrated symbiotic associations of fungi and algae or cyanobacteria.
- Many fungi are parasites, mostly of plants.
- Humans use fungi for food and to make antibiotics.

How are fungi important as decomposers, mutualists, and pathogens?